

Scalability & Future Plan

Date	01 July 2024
Team ID	
Project Name	Human development index (HDI) Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static HDI dataset for prediction.	Predictions are limited to the available training data.	High
2	Supports a limited number of concurrent users.	Supports a limited number of concurrent users.	Medium
3	Basic authentication and security features.	Less suitable for deployment in largescale or production environments.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for small-scale usage.	Deploy on a cloud platform with load balancing to support more concurrent users.
2	Data Storage	Uses CSV files for storing the dataset.	Migrate to a relational database such as MySQL or PostgreSQL for efficient data management.

3	Performance	Prediction is performed on a locally hosted Flask server	Optimize the ML model and deploy on scalable cloud infrastructure for faster response times.
4	Security	Basic input validation and authentication.	Implement HTTPS, stronger authentication, and role-based access control for enhanced security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time HDI and socioeconomic datasets from trusted public sources.	3–6 Months	Improves prediction accuracy and keeps the system up to date.
Phase 3	Add interactive dashboards, trend analysis, and country comparison features.	6–9 Months	Provides deeper insights for researchers and policymakers.
Phase 4	Deploy the application on a cloud platform with advanced security and mobile support.	9–12 Months	Enables wider accessibility, better scalability, and improved user experience.